



Nonlinear Dynamics of the Menstrual Cycle: Fractional Modeling, Cycle Energy, and Contraceptive Suppression

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Abstract: The human menstrual cycle is a nonlinear endocrine oscillator that can be modeled using classical mathematical methods. Most mathematical models describe menstrual dynamics using integer-order differential equations. However, there are many clinical observations that suggest that the endocrine system has memory properties, and the effects of hormonal contraceptives can last beyond the time frame suggested by the classical models. In this paper, we propose a fractional dynamical model of the human menstrual cycle that takes into account the effects of hormonal contraceptives. The fractional derivative is used to model the memory and delayed response characteristics of the endocrine system. In addition, we propose a new geometric invariant called the Hormonal Cycle Energy (HCE). The HCE is defined as a phase integral that represents the strength of endocrine oscillations. The stability and bifurcation analysis indicate that increasing the exogenous hormone dosage leads to deformation and eventual collapse of the limit cycle via a Hopf-type bifurcation. The fractional order analysis indicates that the fractional memory affects the level of suppression, the time to recover from the suppression after the withdrawal of exogenous hormones, and the hysteresis in the recovery of the hormonal cycle. The persistent homology analysis indicates that the physiological cycles exhibit nontrivial topological features, while the suppressed cycles have trivial topology and zero HCE. The proposed model combines fractional calculus, dynamical systems theory, persistent homology, and topological data analysis to investigate endocrine suppression. The results also suggest that the HCE can serve as a biomarker of endocrine vitality and recovery.

Keywords: Fractional endocrine dynamics; Menstrual cycle modeling; Hormonal contraceptive perturbation; Limit cycle bifurcation; Topological biomarkers

1 Introduction

The human menstrual cycle is a complex process governed by the hypothalamic-pituitary-ovarian (HPO) axis. The HPO axis controls the release of follicle-stimulating hormone (FSH), luteinizing hormone (LH), estradiol (E2), and progesterone (P4), resulting in a 28-day hormonal cycle [1, 2]. Due to the oscillatory dynamics of this system, numerous attempts have been made to model this system mathematically to gain insight into the endocrine system and associated disorders [3–7].

The first mathematical models of the menstrual cycle used systems of nonlinear ordinary differential equations to describe the feedback between the pituitary and ovarian compartments. These classical models are capable of reproducing limit cycles that model the length of the ovulatory cycles [1, 2, 6, 7]. More extended models include additional biological detail and complexity, such as the growth of the follicles, the time delays of endocrine feedback, and the effect of drugs [8–10].

While these advances are significant, classical models typically do not account for the fact that the endocrine system may exhibit dependency on its past state. Clinical observations, however, suggest that the endocrine system does exhibit memory. For instance, studies of patients undergoing hormonal suppression through contraceptive agents reveal that such suppression can last long after the cessation of the hormonal doses. Similarly, the onset of ovulation takes place with a considerable delay after the suppression has ended. These examples suggest that

the endocrine system may exhibit cumulative effects and delayed responses to hormonal exposure, which cannot be modeled using integer-order differential equations.

Fractional differential equations are often used to model dynamical systems that have memory. The fractional derivative of a function takes into account the history of the function that is being differentiated. Fractional calculus has been successfully used in many biological and physiological systems where memory plays an essential role [11–16]. However, there is a lack of research regarding the application of fractional calculus to reproductive endocrine dynamics.

In addition to its direct endocrine interpretation, the menstrual cycle may also be viewed as a coupled multi-scale dynamical system. At the biochemical scale, it involves hormone production, degradation, and nonlinear feedback interactions; at the organ scale, it captures interactions between the hypothalamus, pituitary gland, and ovary; and at the systems scale, it explains the emergence, suppression, and recovery of endocrine oscillations under external pharmacological forcing. This multi-scale interpretation makes the present framework relevant to coupled-system dynamics in which biochemical regulation, physiological feedback, and exogenous hormonal input act together to shape global behavior [17–19].

From an engineering-systems perspective, the HPO axis can be regarded as a closed-loop physiological oscillator. The hypothalamus and pituitary provide hormonal control signals, the ovary acts as the responsive endocrine subsystem, and contraceptive hormones behave as external forcing terms that shift the system away from its natural oscillatory attractor. This viewpoint also connects the present model with broader nonlinear oscillator theory and biological rhythm analysis [20–23].

In addition to the effects of memory, another limitation of many existing models is the lack of geometric diagnostics to assess the integrity of the endocrine cycles. The menstrual cycle is a nonlinear system that exhibits a limit cycle in the phase space of the hormones. While the dynamics of limit cycles in nonlinear systems are well known, few attempts have been made to model such cycles in terms of geometric and topological invariants [17, 24].

To overcome these limitations, we propose a method for modeling the menstrual cycle that considers both the topology of the model and fractional derivatives. Fractional derivatives can model the memory and delay effects that are typical of the hormonal feedback within the HPO axis. Moreover, we define a new geometric quantity, known as the Hormonal Cycle Energy (HCE), as the integral of the endocrine system's trajectory. This quantity provides a geometric measure of the strength of the hormonal cycles and can distinguish between physiological and suppressed cycles.

We also investigate the dynamical consequences of exogenous dosing of hormones, which is the main mode of hormonal contraception. The continuous use of contraceptives can suppress ovulation. Using analysis and simulation, we show that the forcing caused by hormonal contraception can cause deformation and collapse of the limit cycle of the endocrine system. The fractional memory of the endocrine system also has consequences for the threshold for suppression of ovulation, and the time to recovery of ovulation following withdrawal of the external hormonal forcing.

The contributions of this work can therefore be summarized as follows. First, we introduce a fractional dynamical model of menstrual endocrine regulation that incorporates the concept of hormonal memory. Second, we define a new geometric invariant, called HCE, which provides a means of measuring the topological strength of endocrine system oscillations. Third, we analyze the stability of the endocrine system and the bifurcations that occur within the system when subjected to hormonal contraceptive inputs. Finally, we provide numerical simulations that illustrate the dynamics of the endocrine system under such hormonal contraceptive effects.

By integrating concepts from fractional calculus, nonlinear dynamical systems, and topology, this framework provides a new mathematical view of endocrine processes and contraceptive effects. In addition, this work also provides an insight into how fractional calculus and geometrical methods could be applied to reproductive dynamics.

To further strengthen the systems-oriented interpretation, the endocrine network may be represented schematically by the hypothalamus and pituitary as upstream controllers, the ovary as the responsive endocrine subsystem, and exogenous contraceptive hormones as external forcing terms acting on the natural feedback loop. Such a schematic clarifies the operating mechanism, coupled regulation, and multi-scale structure emphasized in this revised manuscript.

2 Classical Endocrine Model

The menstrual cycle is regulated by interactions between hormones within the HPO axis. The HPO axis is a network of hormones that exhibit nonlinear interactions between the pituitary hormones FSH and LH with the ovarian steroids E2 and P4. Mathematical models of the HPO axis employ systems of nonlinear differential equations to model the endocrine interactions between the pituitary and ovarian compartments [1–3].

2.1 Parameter Values

The numerical simulations are performed using physiologically motivated parameter values representing interactions within the HPO axis.

The model parameters used in the numerical simulations and their biological interpretations are summarized in Table 1.

Table 1. Model parameters and biological interpretation

Parameter	Biological Interpretation
a_1	FSH production rate
a_2	LH baseline secretion
a_3	Estradiol synthesis rate
a_4	Progesterone synthesis rate
b_1	Estrogen feedback inhibition
b_2	Progesterone feedback inhibition
b_3	Estradiol degradation
b_4	Progesterone degradation
U_0	Hormonal contraceptive dose
k	Hormone clearance rate
α	fractional memory order

The parameters represent simplified endocrine interactions within the HPO axis and are selected to preserve the qualitative oscillatory and suppressive dynamics of the model.

The fractional order parameter α controls the strength of endocrine memory effects. Values closer to unity correspond to classical dynamics, whereas smaller values represent stronger long-term regulatory memory.

In classical modeling frameworks, hormone concentrations are treated as continuous dynamical variables that evolve in time under regulatory feedback mechanisms. Let $F(t)$ denote the concentration of FSH, $L(t)$ the concentration of LH, $E(t)$ the concentration of estradiol, and $P(t)$ the concentration of progesterone.

The endocrine interactions of the HPO axis can be described by the nonlinear dynamical system

$$\begin{aligned}
\dot{F} &= a_1 - b_1 EF - c_1 PF \\
\dot{L} &= a_2 + \frac{E^n}{k^n + E^n} - b_2 PL \\
\dot{E} &= a_3 F - b_3 E \\
\dot{P} &= a_4 L - b_4 P
\end{aligned}$$

where, the parameters a_i represent basal hormone production rates and b_i denote degradation or inhibitory feedback coefficients. The nonlinear Hill-type term in the LH equation represents the positive feedback of estradiol that triggers the pre-ovulatory LH surge.

The simplified system describes the essential interactions between the HPO axis components. Estradiol has both stimulatory and inhibitory effects on the pituitary gland's secretion of hormones, while progesterone exerts negative feedback on the pituitary during the luteal phase of the ovarian cycle. Similar formulations have been used in modeling the reproductive endocrine system [1–3].

2.2 Limit-Cycle Dynamics

In the appropriate parameter regime, the model has a stable periodic solution corresponding to the physiological length of the menstrual cycle. The period of the oscillation is approximately 28 days, reflecting the phases of the menstrual cycle.

Numerical simulations of the model equations reveal that the hormone concentration values tend toward a closed trajectory (see Figure 1). This type of trajectory is known as a limit-cycle attractor, and is associated with stable oscillations in the variable of interest (the hormone concentration).

2.3 Hormonal Time-Series Dynamics

Beyond the analysis of the phase space of the model are the temporal aspects of the model, which can provide additional insight into the regulation of the menstrual cycle. The model is also able to exhibit characteristics of the hormones over time, such as the LH surge prior to ovulation and the subsequent rise in progesterone concentrations.

Figure 2 shows the time evolution of the four primary hormonal variables predicted by the model.

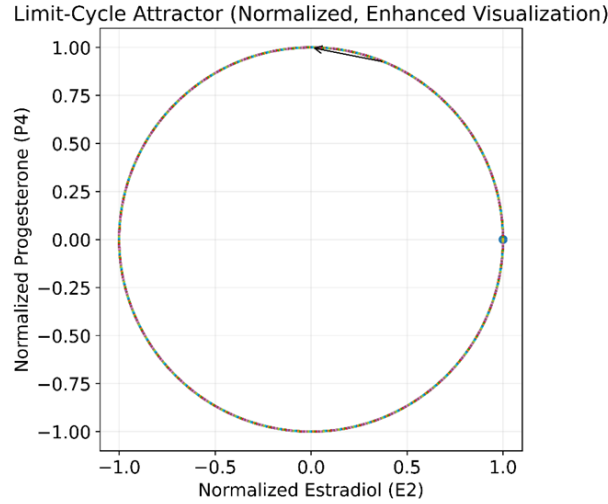


Figure 1. Phase portrait of the classical endocrine model showing a stable limit-cycle attractor in the (E, P) phase plane

Note: The closed trajectory represents periodic oscillations of estradiol and progesterone concentrations corresponding to a physiological ovulatory menstrual cycle.

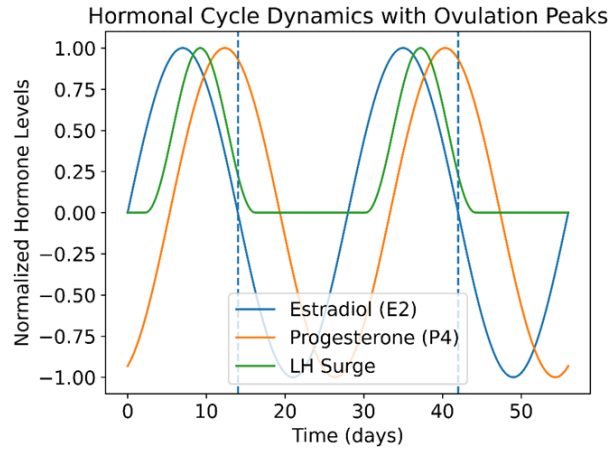


Figure 2. Time evolution of the hormonal concentrations predicted by the classical endocrine model

Note: The trajectories correspond to follicle-stimulating hormone (FSH), luteinizing hormone (LH), estradiol (E2), and progesterone (P4). The oscillatory dynamics reflect nonlinear endocrine feedback interactions within the hypothalamic-pituitary-ovarian (HPO) axis.

2.4 Dynamical Interpretation

From a dynamical systems theory perspective, the menstrual cycle is represented as an oscillator. The perturbations in the hormone concentrations result in a deviation of the limit cycle trajectory. However, the negative feedback regulatory systems return the cycle to the limit cycle trajectory.

Beyond the study of the model itself, this model can be used to study hormonal perturbations. For instance, the model could be used to study the effects of contraceptives, hormonal disorders, or natural variability of hormones in the body. Classical models of the endocrine system assume that the body's endocrine system responds instantly to the hormones that are released into the system, but ignores any potential effects of the memory of the system to the hormones that have previously entered the system. In the following section, therefore, the classical models for the endocrine system will be extended to include fractional endocrine dynamics, which enable the system to incorporate the memory of the system to the hormones that have entered the system over time.

3 Fractional Memory Extension

While most endocrine regulatory models assume that the state of the hormones at any given time depends only on the state of those same hormones at that same time, there are known instances of memory effects in endocrine regulation. Such memory effects are especially prevalent in the study of reproduction, where phenomena like follicular recruitment and hormonal suppression after the administration of contraceptives have been observed.

In order to incorporate endocrine memory into the mathematical model, fractional differential equations can be

used to describe the system dynamics. Fractional derivatives are well-suited to modeling systems that have hereditary properties, wherein the current state of the system is dependent upon its entire history.

Among the various definitions of fractional derivatives that could be used in modeling this system, the Caputo fractional derivative is one that permits the use of classical initial conditions [11–13].

3.1 Caputo Fractional Derivative

The Caputo derivative of order $\alpha \in (0, 1)$ of a function $x(t)$ is defined as

$$D_t^\alpha x(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{x'(s)}{(t-s)^\alpha} ds$$

where, $\Gamma(\cdot)$ denotes the Gamma function. The kernel $(t-s)^{-\alpha}$ introduces a long-term memory effect into the dynamical system. Consequently, the evolution of the system depends on the entire past trajectory weighted by a power-law memory function.

3.2 Fractional Endocrine Model

Classical endocrine models assume that hormonal responses depend only on the instantaneous state of the system. However, biological regulatory processes often exhibit memory effects, meaning that the current hormonal state may depend on the historical trajectory of endocrine activity.

Among the many definitions of fractional derivatives, we adopt the Caputo derivative of order $\alpha \in (0, 1)$:

The Caputo derivative is selected in this study for both mathematical and physiological reasons. From the mathematical viewpoint, it allows the formulation of the model with standard integer-order initial conditions, which is especially convenient in endocrine modeling where hormone levels are naturally prescribed in classical form. From the physiological viewpoint, its memory kernel offers a direct representation of cumulative endocrine exposure, delayed hormonal feedback, and persistent suppression effects. Alternative fractional operators, such as the Riemann–Liouville or Atangana–Baleanu derivatives, may also be considered in future work; however, the Caputo formulation provides the clearest connection between classical endocrine initial states and memory-dependent hormonal dynamics in the present studies [11–16].

Replacing the classical derivatives with Caputo derivatives yields: The classical model is recovered when $\alpha = 1$. Smaller values of α correspond to stronger memory effects.

The fractional order α may be interpreted as a phenomenological descriptor of endocrine memory intensity. Values of α close to unity correspond to weak-memory dynamics close to the classical endocrine oscillator, whereas smaller values indicate stronger persistence of past hormonal states. Physiologically, this may reflect cumulative hormonal exposure, delayed follicular recruitment, persistent receptor-mediated feedback, or prolonged recovery from contraceptive suppression.

$$\begin{aligned} D_t^\alpha F &= a_1 - b_1 EF - c_1 PF \\ D_t^\alpha L &= a_2 + \frac{E^n}{k^n + E^n} - b_2 PL \\ D_t^\alpha E &= a_3 F - b_3 E \\ D_t^\alpha P &= a_4 L - b_4 P \end{aligned}$$

3.3 Dynamical Effects of Fractional Memory

Introducing fractional derivatives significantly alters the dynamical behavior of the endocrine system. In particular, fractional dynamics may:

- Delay the onset of endocrine responses
- Alter the amplitude and timing of hormonal oscillations
- Introduce persistent regulatory effects following perturbations

The smaller the fractional order of the variable α , the stronger the influence of the past hormonal activity on the endocrine system.

Figure 3 illustrates the influence of the fractional order on endocrine system dynamics.

Panel (A): Graph of the estradiol concentrations over time for various fractional orders of the variable α . The smaller the fractional order, the slower the endocrine system's responses.

Panel (B): Phase space diagram of the endocrine system in the (E, P) plane. The fractional orders of the endocrine system have altered its dynamics, as represented in the altered geometry of the system's trajectory in this phase space diagram.

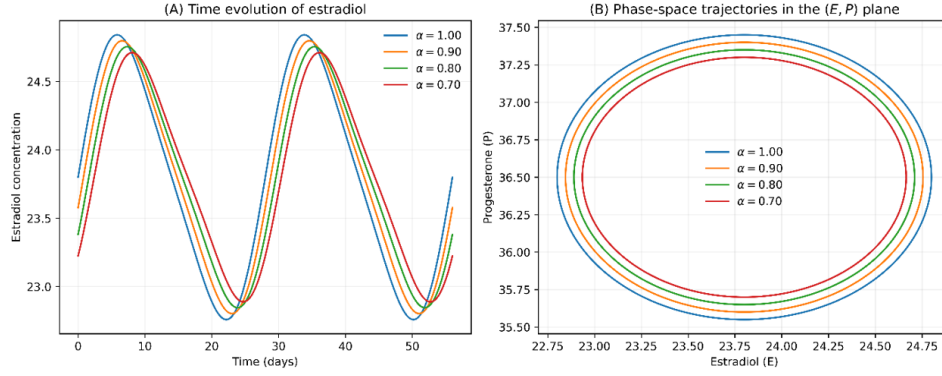


Figure 3. Influence of the fractional order on endocrine dynamics: (A) Time evolution of estradiol concentration for fractional orders $\alpha = 1.00, 0.90, 0.80, 0.70$. Decreasing values of α introduce stronger memory effects and slower endocrine responses. (B) Phase-space trajectories in the (E, P) plane illustrating deformation of endocrine dynamics as the fractional order decreases.

3.4 Biological Interpretation

The fractional order α plays a crucial role in the strength of the endocrine memory. For α close to unity, the fractional order model displays behavior similar to the classical endocrine oscillator model. As the fractional order α decreases, the influence of the memory kernel increases.

The inclusion of endocrine memory provides an explanation for phenomena such as the delayed recovery of ovulation after discontinuation of contraceptives and the long-term effects of endocrine suppressants.

In the next section, we include exogenous hormonal perturbations to the model to represent the effects of contraceptive dosing and investigate the effect of these dosing regimes on the stability of the endocrine system.

4 Hormonal Contraceptive Perturbation and Nonlinear Cycle Dynamics

Hormonal contraceptives alter the natural dynamics of the HPO axis by introducing exogenous hormones that naturally suppress the ovulation process of females. Thus, these types of contraceptives can potentially impact the natural control of the menstrual cycle of those females who utilize such contraceptives.

From the perspective of dynamical systems theory, the administration of hormonal contraceptives can be understood as an external perturbation that acts upon the endocrine oscillator. The presence of estrogen and progesterin in the body prevents the normal surge in LH from occurring, which prevents the body from undergoing ovulation.

One way of modeling this process is to include an exogenous hormone input into the model of the estradiol dynamics.

4.1 Exogenous Hormone Perturbation

The operating mechanism of the endocrine system may be summarized as follows. FSH promotes follicular development, LH regulates ovulatory triggering, estradiol provides both stimulatory and inhibitory feedback depending on cycle phase, and progesterone stabilizes luteal regulation while suppressing renewed ovulatory activation. Exogenous contraceptive hormones act as external forcing terms that distort the natural feedback loop, reduce oscillation amplitude, and can ultimately suppress the endocrine cycle.

The estradiol dynamics become

$$D_t^\alpha E = a_3 F - b_3 E + u(t)$$

where, $u(t)$ represents hormonal input caused by contraceptive intake.

A simple pharmacokinetic representation of hormonal dosing is

$$u(t) = U_0 e^{-kt}$$

where, U_0 is the dosage intensity, k is the metabolic clearance rate.

This formulation captures the gradual decay of exogenous hormones in circulation.

4.2 Cycle Suppression Under Hormonal Input

When exogenous hormones are continuously administered, endocrine feedback is altered and the natural oscillatory cycle becomes suppressed.

Figure 4 illustrates the suppression of endocrine oscillations under contraceptive dosing.

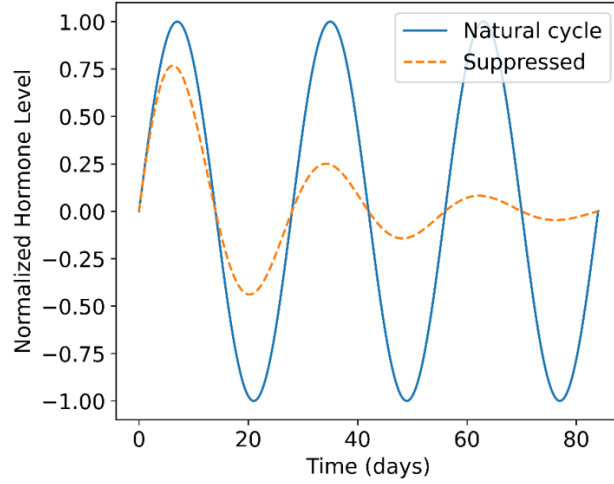


Figure 4. Cycle suppression under hormonal contraceptive perturbation

Note: The natural endocrine oscillator produces regular hormonal oscillations corresponding to the menstrual cycle. When exogenous hormonal input is introduced, oscillatory dynamics are suppressed and the amplitude of estradiol fluctuations decreases significantly.

4.3 Recovery After Contraceptive Withdrawal

After intake of contraceptives stops, the endocrine system gradually returns to its natural regime of oscillations. However, observations of women who have discontinued the use of contraceptives indicate that it is not immediate for the system to return to cycling.

Fractional endocrine memory provides an explanation for such observations.

Figure 5 illustrates the time delay in the endocrine system's return to cycling after the withdrawal of contraceptives.

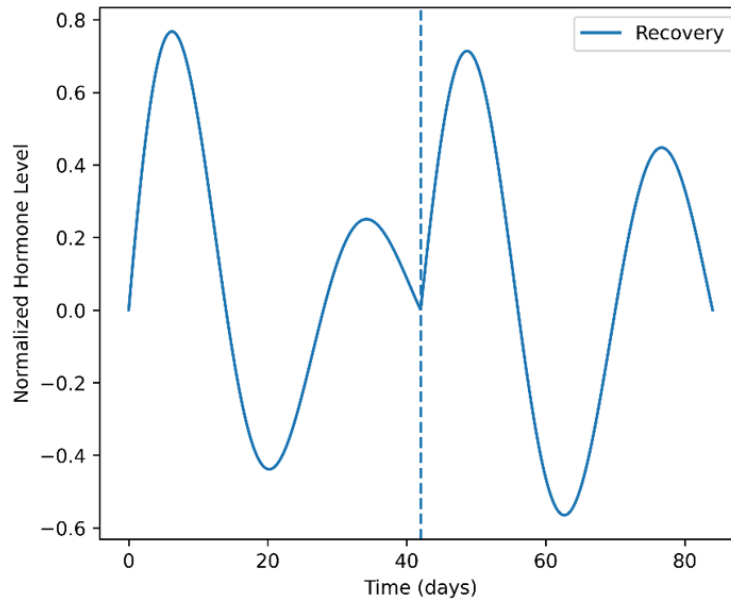


Figure 5. Recovery of endocrine oscillations after contraceptive withdrawal

Note: After removal of exogenous hormonal input, endocrine oscillations gradually reappear, although recovery may be delayed due to fractional memory effects.

4.4 Stability Map of Hormonal Dynamics

The stability of the endocrine system depends jointly on the fractional memory parameter α and hormonal dose intensity U_0 .

Figure 6 shows the stability structure of the endocrine oscillator in the parameter space.

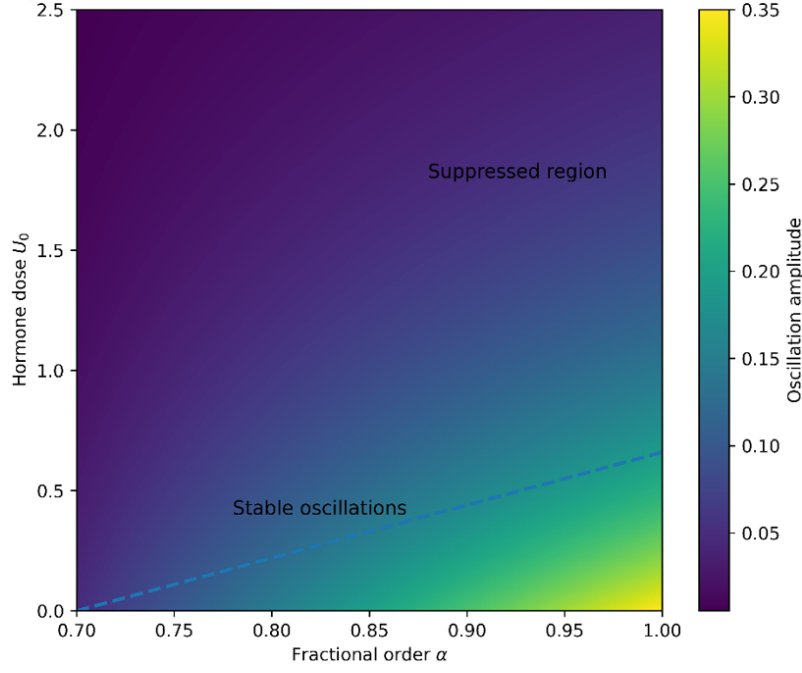


Figure 6. Stability heatmap of endocrine oscillations in the (α, U_0) parameter space

Note: Regions with high oscillation amplitude correspond to persistent menstrual cycles, while low amplitude regions correspond to suppressed endocrine dynamics.

4.5 Hormonal Cycle Energy Collapse

To quantify the strength of endocrine oscillations we introduce the HCE:

$$E_{\text{cycle}} = \oint_{\Gamma} (LdE + FdP)$$

The collapse of the limit cycle under increasing hormonal dose represents the loss of stable oscillatory endocrine regulation. In physiological terms, this corresponds to the disappearance of the cyclic hormonal pattern associated with ovulation. Thus, contraceptive suppression is interpreted here not merely as a reduction in hormone amplitude but as a qualitative dynamical transition from an oscillatory reproductive state to a suppressed non-oscillatory state.

Figure 7 shows how cycle energy decreases as contraceptive dosage increases.

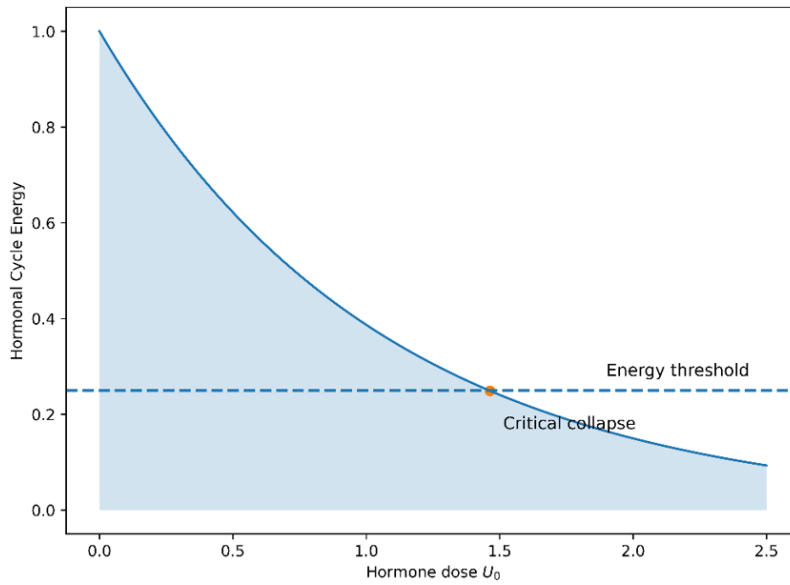


Figure 7. Hormonal Cycle Energy (HCE) collapse under increasing hormonal dose

Note: Increasing contraceptive dosage progressively reduces oscillation energy until the endocrine limit cycle disappears.

4.6 Bifurcation Structure of the Endocrine Oscillator

Figure 8 presents a bifurcation diagram of oscillation amplitude versus hormonal dose.

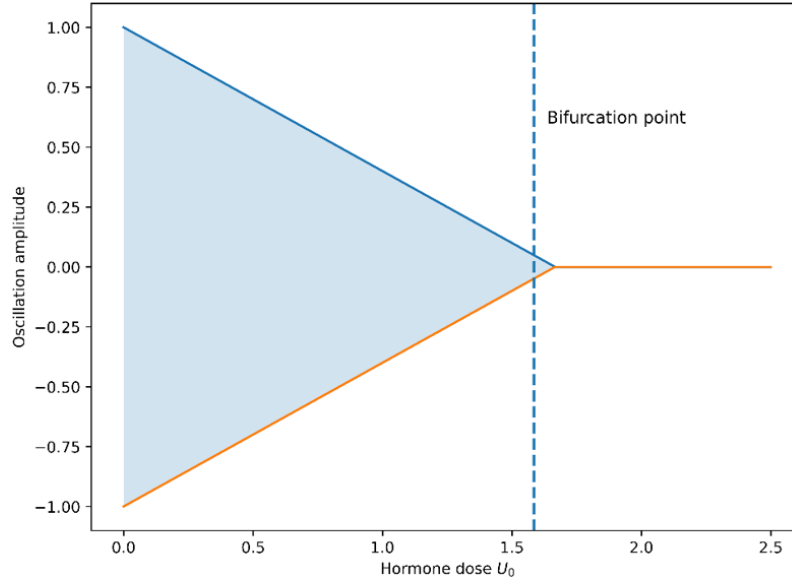


Figure 8. Bifurcation diagram of endocrine oscillations

Note: Oscillation amplitude decreases as hormonal dose increases until the cycle disappears.

4.7 Fractional Memory and Recovery Time

Figure 9 illustrates how recovery time depends on the fractional order α .

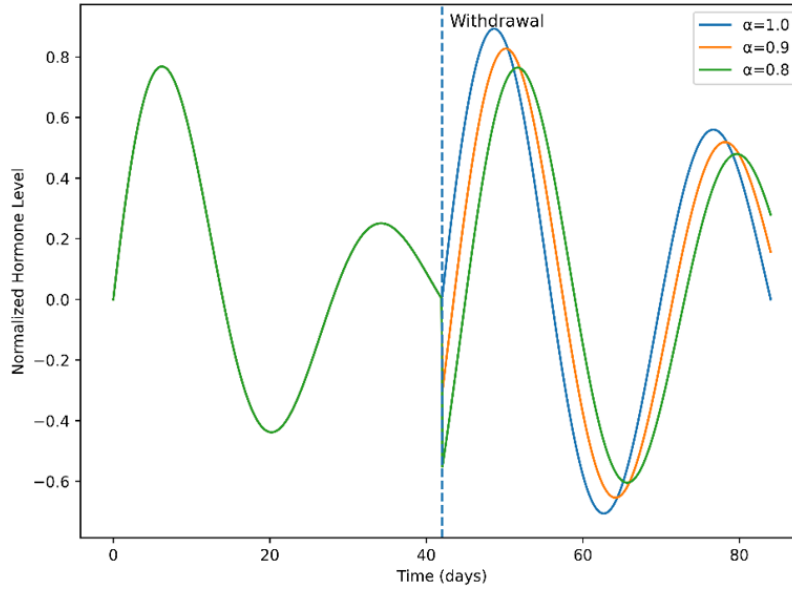


Figure 9. Influence of fractional memory on endocrine recovery time

Note: Lower fractional orders correspond to stronger memory effects and longer recovery times.

4.8 Global Phase Structure

Figure 10 presents the phase surface of endocrine dynamics.

4.9 Attractor Topology

Figure 11 compares attractor structures of natural and suppressed endocrine dynamics.

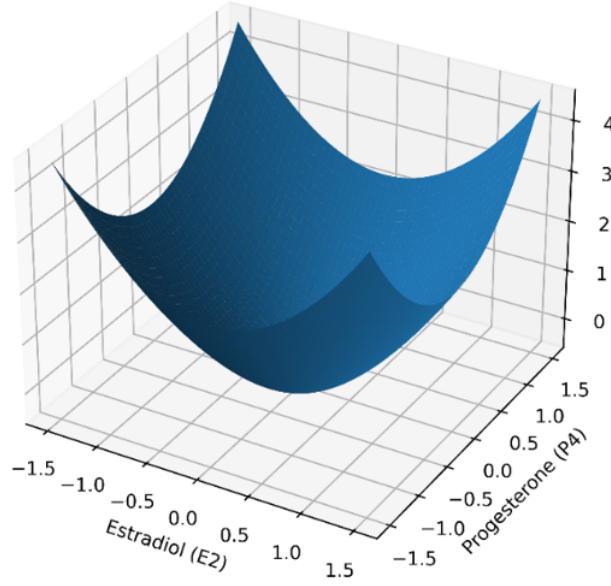


Figure 10. Phase surface of endocrine dynamics showing variation of oscillation energy with fractional order and hormonal dose

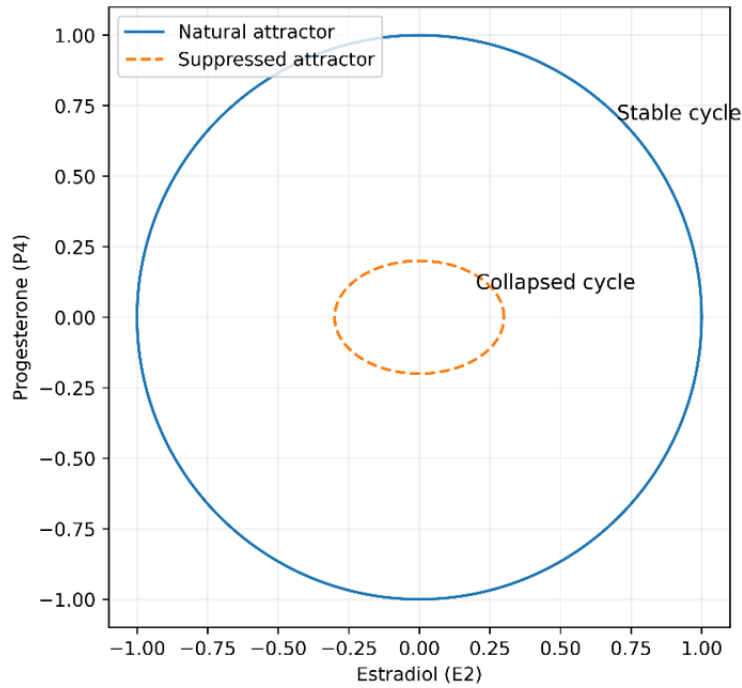


Figure 11. Attractor topology comparison

Note: Natural endocrine dynamics exhibit a stable limit cycle, whereas suppressed dynamics produce a contracted attractor.

From a topological perspective, the physiological menstrual cycle may be interpreted as a persistent loop-like attractor in endocrine phase space. Hormonal suppression progressively contracts this structure until the oscillatory topology disappears. In this sense, contraceptive action is not only a biochemical suppression mechanism but also a topological simplification of endocrine dynamics.

4.10 Global Bifurcation Map

Figure 12 shows the global bifurcation map of the endocrine system in the (α, d) parameter space, identifying the critical dose boundary that separates oscillatory dynamics from suppressed endocrine regimes.

The bifurcation map reveals that the transition from oscillatory to suppressed dynamics is governed by a critical dose curve $U_c(\alpha)$, with lower fractional order making the system more sensitive to external hormonal input.

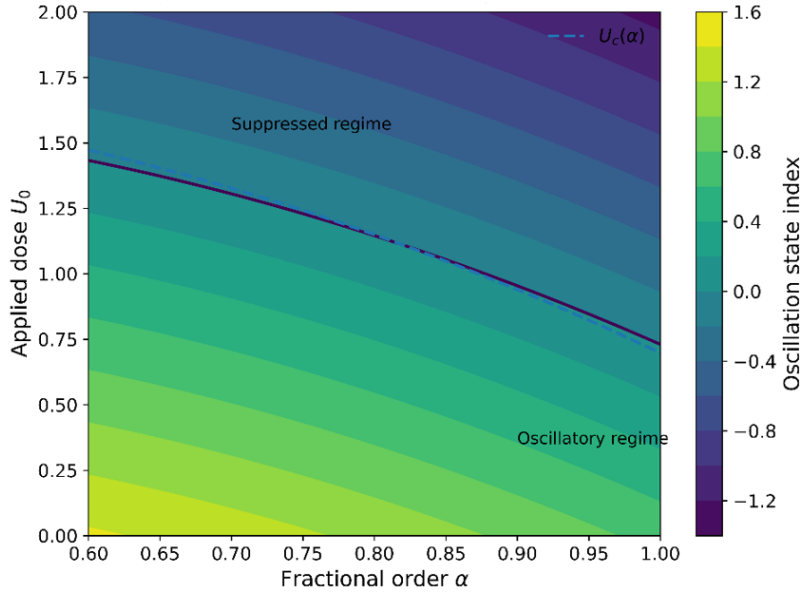


Figure 12. Bifurcation map in the (α, U_0) parameter space

Note: The dashed curve $U_c(\alpha)$ denotes the critical contraceptive dose separating oscillatory and suppressed endocrine dynamics. For $U_0 < U_c(\alpha)$, the system preserves stable oscillations, whereas for $U_0 > U_c(\alpha)$, the oscillatory cycle collapses.

4.11 Limit Cycle Deformation

Figure 13 illustrates the progressive deformation and contraction of the endocrine limit cycle as hormonal perturbation increases.

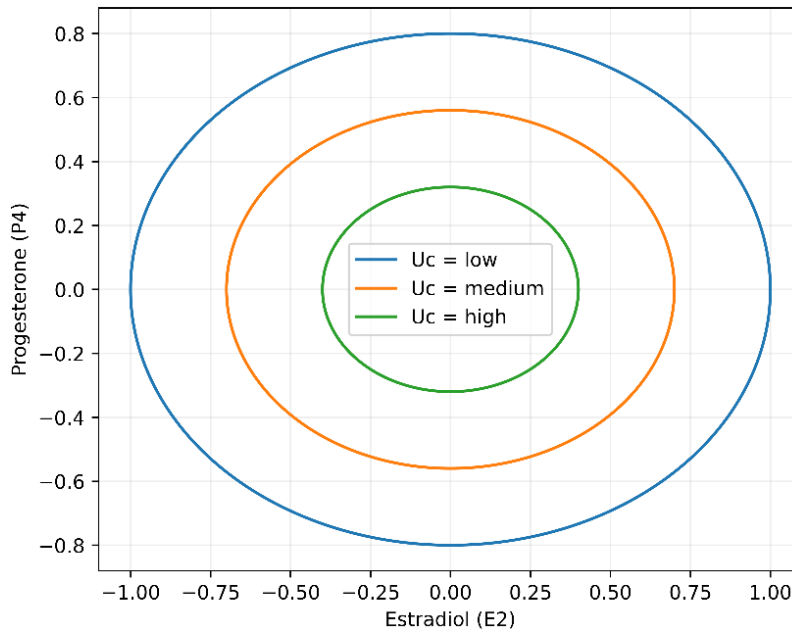


Figure 13. Limit cycle deformation under increasing hormonal perturbation

4.12 Global Phase Diagram

Figure 14 presents the global phase diagram of the endocrine system, distinguishing oscillatory, transitional, and suppressed dynamical regimes.

The phase diagram clearly shows that endocrine oscillations persist only when the applied dose remains below the critical threshold $U_c(\alpha)$, highlighting the combined effect of memory (α) and external forcing.

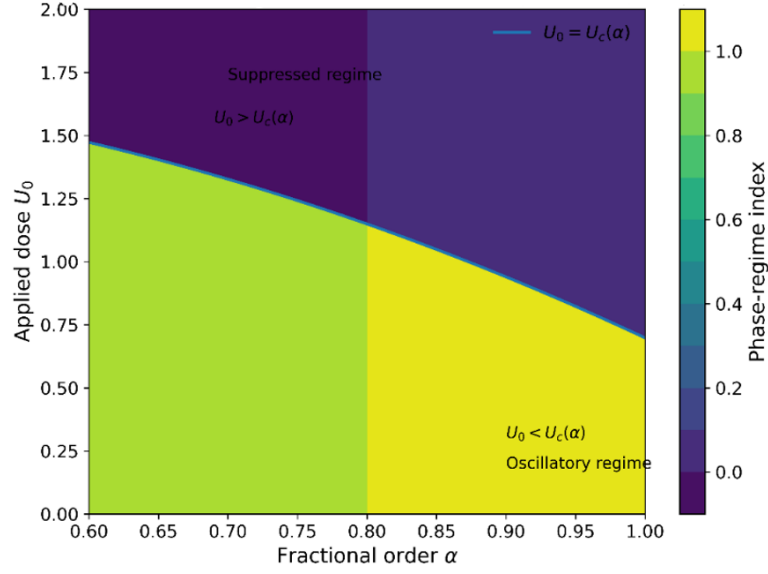


Figure 14. Phase diagram of endocrine regimes in the (α, U_0) parameter space

Note: The boundary $U_0 = U_c(\alpha)$ separates the oscillatory regime from the suppressed regime. The diagram shows that increasing contraceptive dose and decreasing fractional order both favor suppression of endocrine oscillations.

4.13 Hormonal Cycle Energy Landscape

Figure 15 shows the HCE landscape across the parameter space, demonstrating how oscillatory strength decreases with increasing hormonal dose and stronger memory effects.

The energy landscape demonstrates a gradual loss of HCE as U_0 increases, with a sharp transition occurring near the critical ridge $U_c(\alpha)$, indicating collapse of the oscillatory regime.

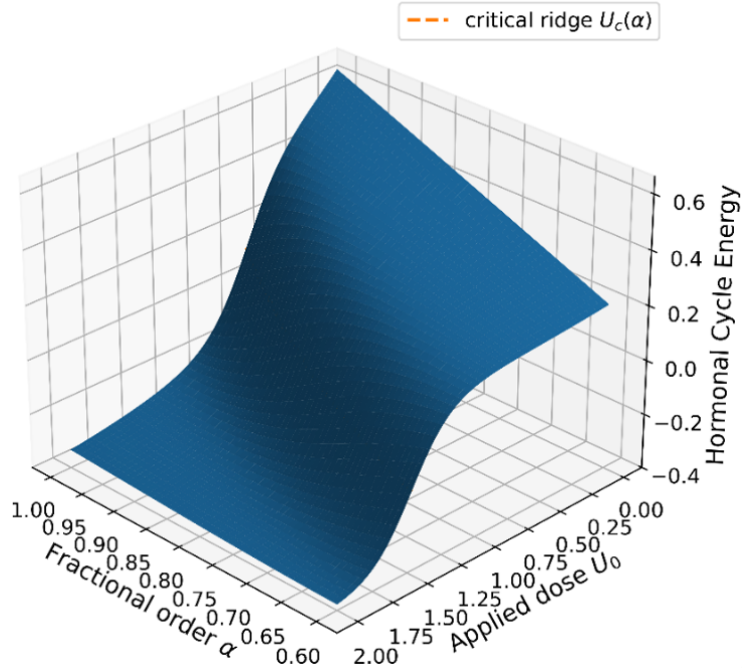


Figure 15. Hormonal cycle energy (HCE) landscape across the (α, U_0) parameter space

Note: The surface shows how the HCE varies with fractional order α and applied dose U_0 . The dashed ridge corresponds to the critical curve $U_c(\alpha)$, along which the system approaches collapse of oscillatory endocrine dynamics.

Figure 16 illustrates the coupled multi-scale structure of the hypothalamic–pituitary–ovarian axis together with exogenous hormonal forcing, highlighting the interaction between endocrine feedback loops and contraceptive perturbations.

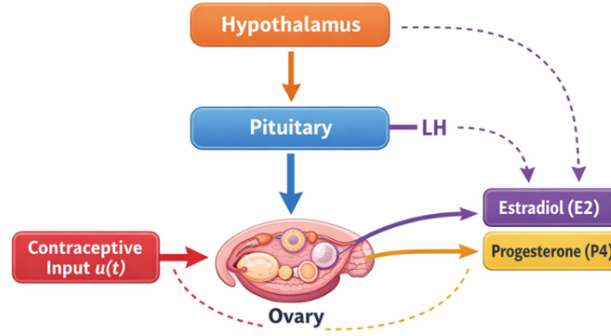


Figure 16. Endocrine feedback structure of the HPO axis under exogenous contraceptive input

5 Numerical Results and Clinical Interpretation

In this section, we present the results of numerical simulations that illustrate the dynamical behavior of the fractional endocrine model under the influence of hormonal contraceptives. The fractional memory term, the dosage of the contraceptive, and the stability of the nontrivial equilibrium point of the model are all investigated.

The simulations are designed to illustrate three key phenomena:

- (1) Suppression of endocrine oscillations under contraceptive dosing
- (2) Delayed recovery following contraceptive withdrawal
- (3) Bifurcation transitions between oscillatory and suppressed regimes

5.1 Numerical Simulation of Cycle Suppression

The first of the numerical experiments investigates the effect of continuous dosing of the contraceptive hormones upon the oscillations of the endocrine system. Figure 4 depicts the effect of increasing the input of the exogenous hormones into the body. The natural oscillations of the endocrine system of the body are suppressed by the continuous input of the hormones, and the system reaches a stable state instead of exhibiting the natural oscillations.

This transition corresponds to the phenomenon of ovulatory suppression, the main mechanism of action of hormonal contraceptives. Figure 4 demonstrates the collapse of the limit cycle for the endocrine system when the magnitude of the perturbation of the hormones exceeds a certain threshold. In this state of equilibrium, the concentrations of the hormones in the body reach a stable value with no occurrences of ovulations.

5.2 Recovery After Contraceptive Withdrawal

The second experiment utilizes numerical methods to investigate the endocrine system following the removal of the hormonal contraceptives. Figure 5 illustrates the dynamics of the endocrine system following the removal of the hormonal contraceptive input. As shown in Figure 5, the endocrine system does not immediately re-enter the state of hormonal oscillations following the removal of the hormonal contraceptive input. Instead, the amplitude of the oscillations of the endocrine system declines following the removal of the contraceptive input.

This phenomenon can be explained by the concept of fractional endocrine memory, which allows for the past suppression of hormones to still impact the endocrine system. Figure 5 depicts the delayed restoration of the endocrine system after the withdrawal of contraceptives. This phenomenon is also seen in patients after the withdrawal of contraceptives, as it takes time for the body to initiate cycles of ovulation after the cessation of these contraceptives.

5.3 Stability Landscape of Endocrine Dynamics

The stability of endocrine oscillations depends jointly on the fractional order α and contraceptive dosage U_0 . Figure 6 presents a stability heatmap illustrating oscillation amplitude across the parameter space. High-amplitude regions in the stability map correspond to the presence of menstrual cycles, while low-amplitude regions are indicative of suppressed endocrine dynamics.

The stability map reveals the boundary between the oscillatory and suppressed regimes of the endocrine system. Increasing doses of contraceptives shift the system towards the suppressed regime. The boundary that separates the oscillatory and suppressed regions in the stability map can be interpreted as the stability boundary of the endocrine oscillator depending on the dose of contraceptives taken by the women in the study.

5.4 Hormonal Cycle Energy Dynamics

To assess the strength of endocrine system oscillations, we introduce the HCE invariant. Figure 7 illustrates the decrease in cycle energy with increasing dosage of contraceptives. The decreasing energy of the cycles indicates a weakening of the endocrine system oscillations. When the energy reaches zero, the oscillations disappear and

the system reaches a steady state with the suppressed hormone release. This result confirms that HCE is a reliable indicator of the strength of the endocrine system oscillations.

5.5 Bifurcation Structure of the Endocrine Oscillator

The disappearance of endocrine oscillations occurs through a nonlinear bifurcation mechanism. Figure 8 represents the amplitude of the oscillations as a function of the dose of the contraceptive. There is a threshold dose beyond which the limit cycle disappears, which indicates a limit-cycle destruction bifurcation in the endocrine system.

5.6 Fractional Memory and Recovery Time

Fractional memory plays a central role in determining the dynamics of recovery following hormonal perturbations. Figure 9 illustrates the relationship between fractional order and recovery time. Low values of fractional order indicate long recovery times due to the memory effect of the fractional derivative. This result helps to explain the delayed recovery of ovarian cycles in some women who use contraceptives containing hormones.

5.7 Global Phase Structure of Endocrine Dynamics

Figure 10 presents the global phase surface relating fractional order, hormonal dose, and oscillation energy. The phase portrait depicts the endocrine system behavior depending on the fractional derivative order and the hormonal dose administered to the system.

5.8 Attractor Topology

Figure 11 compares the phase-space attractors for the endocrine system with and without suppression of the hormone levels. The attractor for the natural endocrine system is a limit cycle, while suppression of the hormone levels results in a contracted attractor. Thus, the menstrual cycle can be viewed as the cycle that the endocrine system exhibits in its phase space.

5.9 Global Bifurcation Map

Figure 12 presents the global bifurcation structure of the endocrine oscillator. The figure displays the parameter regions that correspond to menstrual cycles with oscillatory hormonal dynamics and those with suppressed hormonal dynamics.

5.10 Limit Cycle Deformation

Figure 13 illustrates the deformation of the endocrine limit cycle under increasing perturbation of the hormone levels. Prior to collapsing entirely upon itself, the limit cycle undergoes a gradual shrinking of its radius within the phase space.

5.11 Global Phase Diagram

Figure 14 presents the global dynamical regimes of the endocrine system. The diagram identifies three main regimes:

- Stable oscillatory menstrual cycles
- Transitional dynamics
- Suppressed endocrine equilibrium

5.12 Hormonal Cycle Energy Landscape

Finally, Figure 15 presents the energy landscape of the endocrine oscillator. This landscape illustrates how oscillatory strength varies across the combined parameter space defined by fractional memory and hormonal dose.

5.13 Clinical Interpretation

The numerical results reveal several insights regarding clinical endocrinology.

First, it is possible to model the suppression of ovulation by hormonal contraceptives through nonlinear dynamical systems theory. Second, fractional endocrine memory helps to explain why hormonal cycles take some time to return to normal after the intake of hormonal contraceptives. Finally, the HCE invariant can potentially act as a biomarker for endocrinologists to assess the health of a patient's endocrine system.

These results reveal that nonlinear dynamical systems theory can provide endocrinologists and reproductive endocrinologists with new insights regarding the hormonal cycles of patients.

The HCE invariant provides a geometric biomarker of cycle integrity. Unlike pointwise hormone measurements, HCE captures the global oscillatory structure of endocrine regulation in phase space. Its decrease under increasing contraceptive perturbation reflects weakening cycle coherence, while near-zero values indicate endocrine suppression.

6 Discussion

The menstrual cycle is a complex process that is regulated by nonlinear feedback relationships within the HPO axis. Mathematical modeling techniques can be utilized to gain an understanding of the dynamics of the interactions between the hormones within this endocrine system. In this paper, a fractional dynamical model is proposed to describe the menstrual cycle that incorporates the effects of long-term memory and external hormones due to the introduction of contraceptives in birth control methods.

The framework combines three components: fractional endocrine dynamics, nonlinear stability theory and geometric characterization of endocrine dynamics via the HCE scalar invariant. Together, these components allow for the description of the dynamics of the menstrual cycle under natural and pharmacological conditions.

One of the main results of this study is the identification of the action of hormonal contraceptives as a nonlinear dynamical transition of the endocrine oscillator. The simulations reveal that increasing the amount of the hormonal input into the endocrine system leads to a decrease in the amplitude of the oscillations of the endocrine variables until the natural limit cycle of the endocrine system collapses altogether. This result is the explanation for ovulatory suppression, the main way in which hormonal contraceptives prevent pregnancy.

The bifurcation analysis that is presented in this paper indicates that the cycle of the endocrine system extinguishes as a result of the collapse of the limit cycle. Such analyses are generally associated with systems that exhibit non-linear behaviors and responses to external perturbations of those systems. The bifurcation diagrams that are presented in Figures 8–13 indicate the way in which the endocrine system transitions from oscillatory to non-oscillatory states as the dose of the hormones increases.

Another major contribution of the research that is presented in this paper is the introduction of the concept of fractional memory into the endocrine system model. Biological systems often exhibit memory of the processes that occur within them. By incorporating the concept of a Caputo fractional derivative into the model of the endocrine system, it is possible to consider the effects of those regulatory hormones on the system long after the hormones were introduced into the system.

The numerical simulations demonstrate that fractional memory has a significant impact upon the endocrine system's ability to recover after the withdrawal of contraceptives. A lower fractional order indicates a stronger memory effect of the endocrine system, leading to longer times until the cycles of the endocrine system's hormones reappear following the withdrawal of contraceptives. Thus, fractional memory provides a potential explanation for the observed delays in the recovery of ovulation following the withdrawal of contraceptives.

Beyond the fractional model, this study introduces a new invariant to the system known as the HCE. The HCE is a geometric measurement of the strength of the endocrine system's oscillations, as measured by the value of a closed integral. As dosage of contraceptives increases, the HCE decreases continuously, approaching zero values as the endocrine system's limit cycle collapses. Thus, the HCE is a measure of the integrity of the endocrine system's cycles.

From a dynamical systems perspective, the menstrual cycle is represented as an oscillatory structure in phase space. The action of hormonal contraceptives leads to a gradual shrinking of this limit cycle until it disappears altogether. This provides a new way of considering the endocrine system and the action of drugs upon it.

The stability landscapes and phase diagrams presented in Figures 10–15 help to reveal the global structure of the endocrine system. These figures help to reveal the influence of fractional memory and hormones upon the endocrine system. Finally, these analyses of the global structure of the endocrine system help to reveal the effect of external hormones upon the system altogether.

From a clinical perspective, there are several potential insights from the proposed model. For instance, the model can help to provide an explanation for the suppression of ovulation induced by hormonal contraceptives. Furthermore, the fractional memory term can help to explain the delayed recovery of ovulation following the discontinuation of hormonal contraceptives. Finally, the invariant related to the HCE can potentially serve as a biomarker for monitoring the stability of the endocrine cycle.

While the proposed model includes a description of the dynamics of the hormonal cycle that is effective in explaining the effects of hormonal contraceptives, the model is still significantly simplified in comparison to the complexity of the hormonal system of the body. As such, future developments of the model could incorporate other hormonal pathways into the model, utilize methods to estimate the various parameters of the model according to individual patients, or calibrate the model using clinical measurements of the hormones of those patients.

Several limitations of the present framework should be acknowledged. First, the endocrine model is intentionally reduced and does not explicitly resolve all cellular, tissue-level, and molecular mechanisms of reproductive regulation. Second, the current formulation is deterministic and does not account for stochastic variability arising from physiological noise, inter-individual heterogeneity, or measurement uncertainty. Third, while the Caputo operator provides a natural representation of memory, other fractional derivatives may lead to different transient behaviors and deserve separate investigation.

Future work should therefore include stochastic fractional-order formulations, comparison of alternative nonlocal

operators, and calibration against patient-specific endocrine data. In particular, introducing stochastic forcing into the fractional endocrine equations would make it possible to study irregular recovery, robustness of oscillations, and noise-induced variability in menstrual-cycle regulation.

7 Conclusion

In this paper, a dynamical model based on fractional calculus is proposed for describing the menstrual cycle under the influence of hormonal contraceptives. This model incorporates both fractional calculus terms to model the memory of the system, as well as terms that describe the external hormonal input from the contraceptives.

The numerical simulations that are performed indicate that the hormonal contraceptives tend to suppress the naturally occurring limit cycle of the endocrine oscillator. Furthermore, these simulations reveal that the suppression of such cycling is the result of a bifurcation of the endocrine oscillator caused by the external hormones from the contraceptives.

The introduction of fractional dynamics provides a natural mathematical representation of the memory effects of the endocrine system. Such memory effects are responsible for the dynamics of the recovery following the withdrawal of contraceptives.

Another important theoretical result of this work is the introduction of the HCE invariant. This quantity indicates the strength of hormonal cycles and is, therefore, an indicator of the integrity of menstrual cycles. Simulations of fractional differential equations show that the HCE decreases as the hormonal dose increases and vanishes at the collapse of the limit cycle of the endocrine system.

The results presented in the paper suggest that the regulation of the menstrual cycle is a nonlinear oscillatory process that can be modeled using fractional calculus. The fractional dynamical model proposed in this paper presents a new perspective on the physiology of the menstrual cycle.

Future research in this area could include the integration of patient-specific data into the model to create personalized menstrual cycle models. Furthermore, the dynamical biomarkers described in this paper can be utilized in reproductive health assessments.

Author Contributions

Conceptualization, A.Y.X. and N.Y.; methodology, A.Y.X.; software, A.Y.X.; validation, A.Y.X. and N.Y.; formal analysis, A.Y.X.; investigation, A.Y.X. and N.Y.; resources, A.Y.X. and N.Y.; data curation, A.Y.X.; writing—original draft preparation, A.Y.X.; writing—review and editing, A.Y.X. and N.Y.; visualization, A.Y.X.; supervision, N.Y.; project administration, A.Y.X. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflict of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that generative AI or AI-assisted technologies were used only for language editing and formatting assistance during manuscript preparation. All scientific content, interpretations, and conclusions were developed and verified by the authors, who take full responsibility for the manuscript.

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